

2D Navier–Stokes recurrent adversarial attack

(20 frames: $x_0, \dots, x_{19}$; input window 0–9, recurrent prediction 10–19)

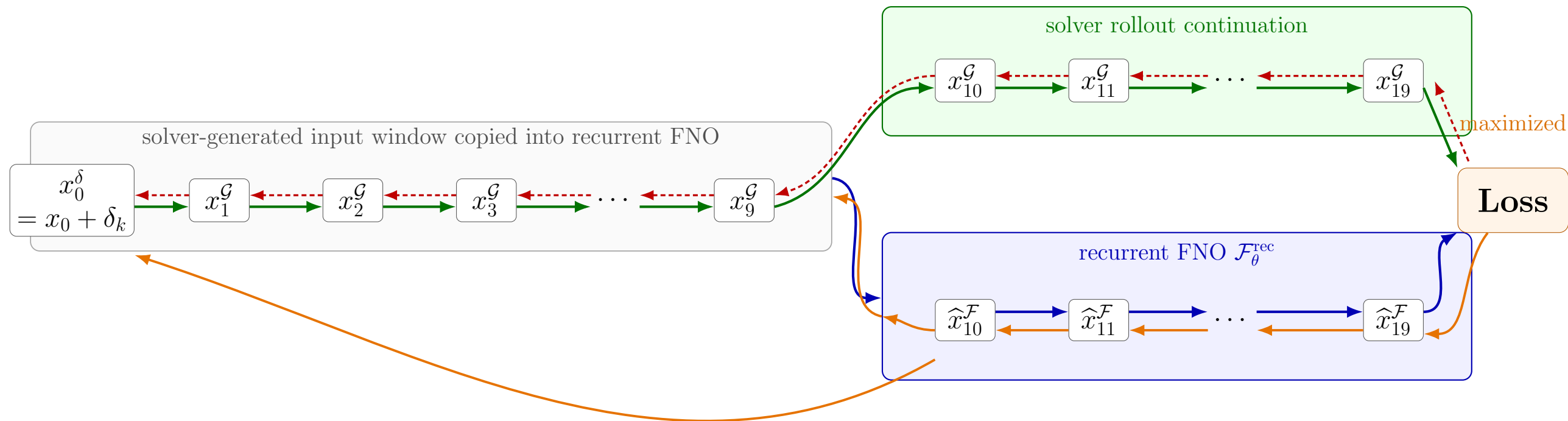
→ $\mathcal{G}$ forward
 → $\mathcal{F}_\theta$ forward
 → $\mathcal{F}_\theta$ backward
 - - - - -> $\mathcal{G}$ backward (detach: no backprop)

$$x_0^\delta = x_0 + \delta_k, \quad x_j^\mathcal{G} = \mathcal{G}_j(x_0^\delta), \quad j = 1, \dots, 19$$

$$[\hat{x}_{10}^\mathcal{F}, \dots, \hat{x}_{19}^\mathcal{F}] = \mathcal{F}_\theta^{\text{rec}}([x_0^\delta, x_1^\mathcal{G}, \dots, x_9^\mathcal{G}]), \quad \mathcal{L}_{\text{adv}}(\delta_k) = \|\hat{x}_{19}^\mathcal{F} - x_{19}^\mathcal{G}\|_2^2$$

$\text{mode_spec} = (c_1, \dots, c_9, c_{19}), \quad c_j \in \{A, D, W\}$

A: dictionary/approx.; D: detached solver; W: solver backprop



$$\nabla_{\delta_k} \mathcal{L}_{\text{adv}} = D\mathcal{G}_{0:9}^* D(\mathcal{F}_\theta^{\text{rec}})^* \nabla_{\hat{x}_{19}^\mathcal{F}} \ell + D\mathcal{G}_{19}^* \nabla_{x_{19}^\mathcal{G}} \ell$$